

Lab2

6. Create a view for student names with their subjects' names which they will study.

Query Query History

```
1 CREATE VIEW student_view AS
2 SELECT student.e_name, subject.sub_name
3 FROM student
4 JOIN stu_sub ON student.id = stu_sub.stu_id
5 JOIN subject ON stu_sub.sub_id = subject.id;
6
7 SELECT * FROM student_view;
8
```

Data Output Messages Notifications

	e_name character varying (100)	sub_name character varying (100)
1	Ahmed	python programming
2	Sara	web developer
3	Omar	Database fundamental

7. Create a view for tracks names and the subjects which belong to it.

Query Query History

```
1 CREATE VIEW track_subjects_view AS
2 SELECT t.track_name, s.sub_name
3 FROM track_sub ts
4 JOIN track t ON ts.track_id = t.id
5 JOIN subject s ON ts.sub_id = s.id;
6
7 SELECT * FROM track_subjects_view;
8
```

Data Output Messages Notifications

	track_name character varying (100)	sub_name character varying (100)
1	python	python programming
2	javascript	web developer
3	database	Database fundamental

1-Write a query to find out which subjects are not associated with any track.

The screenshot shows a database query editor with a 'Query' tab selected. The query is as follows:

```
1 SELECT sub_name FROM subject
2 WHERE id NOT IN (
3     SELECT sub_id FROM track_sub
4 );
5
```

Below the query editor, there are tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Data Output' tab is active, showing a table with one column:

sub_name
character varying (100)

2-Display name and age of each students

The screenshot shows a database query editor with a 'Query' tab selected. The query is as follows:

```
1 SELECT e_name, AGE(birth_date) AS age
2 FROM student;
3
```

Below the query editor, there are tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Data Output' tab is active, showing a table with two columns:

	e_name character varying (100)	age interval
1	Ahmed	25 years 3 mons 17 days
2	Sara	34 years 11 mons 12 days
3	Omar	29 years 8 mons 22 days

3-Display the name of students with their rounded score in each subject

Query		Query History	
1	SELECT	e_name, sub_name ,ROUND(grade)	FROM grades
2	JOIN	student ON grades.stu_id = student.id	
3	JOIN	subject ON grades.sub_id = subject.id;	
4			
Data Output		Messages	
		Notifications	
	e_name character varying (100)	sub_name character varying (100)	round double precision
1	Ahmed	python programming	85
2	Sara	web developer	90
3	Omar	Database fundamental	70

4-Display the name of students with the year of Birthdate

Query		Query History	
1	SELECT	e_name, DATE_PART('year', birth_date) as bir	
2	FROM	student;	
Data Output		Messages	
		Notifications	
	e_name character varying (100)	birth_year double precision	
1	Ahmed	2000	
2	Sara	1990	
3	Omar	1995	

5-Add new exam result, in date column use NOW() function

Query		Query History
1	▼	INSERT INTO exam (date)
2		VALUES (NOW());
3		select * from exam;
Data Output		Messages Notifications
≡+ 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈 SQL		
	id	date
	[PK] integer	date
1	1	2024-12-01
2	2	2024-12-05
3	3	2024-12-10
4	4	2025-05-01

6-Write a query to calculate the average grade obtained by a specific student across all exams.

Query		Query History	Execute script F5
1	▼	SELECT AVG(grade) AS average_grade	
2		FROM grades	
3		WHERE stu_id = 1;	
Data Output		Messages Notifications	
≡+ 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈 SQL			
	average_grade		
	numeric		
1	85.0000000000000000		

7-Write a query to replace all occurrences of 'gmail.com' in email addresses with 'iti.com'.

Query

Query History

1

▼

UPDATE

student

2

SET email = replace(email, 'gmail.com', 'iti.com')

3

WHERE email LIKE '%gmail.com';

4

select * from student;

Data Output

Messages

Notifications

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▼

📋

▼

🗑

🔄

📶

⏏

SQL

	Id [PK] integer	e_name character varying (100)	email character varying (100)	address text	track_id integer	birth_date date	gender gender_type
1	1	Ahmed	ahmed@iti.com	Cairo	1	2000-01-15	Male
2	2	Sara	sara@iti.com	Assiut	2	1990-05-20	Female
3	3	Omar	omar@iti.com	Giza	3	1995-08-10	Male

8-Write a query to calculate the difference in days between the current date and each exam date.

Query		Query History		
1		SELECT id, date,		
2		DATE_PART('day', NOW() - date) AS days_difference		
3		FROM exam;		
Data Output		Messages		
		Notifications		
		SQL		
	id [PK] integer	date date	days_difference double precision	
1	1	2024-12-01	151	
2	2	2024-12-05	147	
3	3	2024-12-10	142	
4	4	2025-05-01	0	

9-Write a query to check if each student's email address ends with '.com'.

Query Query History

```

1 SELECT id, e_name, email,
2     CASE
3         WHEN email LIKE '%.com' THEN 'Yes'
4         ELSE 'No'
5     END AS ends_with_com
6 FROM student;

```

Data Output Messages Notifications

	id [PK] integer	e_name character varying (100)	email character varying (100)	ends_with_com text
1	1	Ahmed	ahmed@iti.com	Yes
2	2	Sara	sara@iti.com	Yes
3	3	Omar	omar@iti.com	Yes

10-Display each exam date like 'MM/DD/YYYY'.

Query Query History

```

1 SELECT id,
2     TO_CHAR(date, 'MM/DD/YYYY') AS formatted_date
3 FROM exam;

```

Data Output Messages Notifications

	id [PK] integer	formatted_date text
1	1	12/01/2024
2	2	12/05/2024
3	3	12/10/2024
4	4	05/01/2025

